Siddhartha Goswami

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About

Mathematics & Computing undergraduate at IIT Roorkee focused on **Generative AI, Agentic Systems, RAG, and Machine Learning**. Experienced in building end-to-end AI products across **enterprise search, intelligent automation, and personalization**.

Education

Indian Institute of Technology Roorkee — Roorkee, India
BS-MS Mathematics and Computing — *Aug 2023 – May 2028*

Experience

- **AI Engineering Intern** — Apr 2026 – Jun 2026
  Hiring Bazaar AI
  - Built an end-to-end **resume parsing and hybrid retrieval pipeline**, combining **BM25 lexical search** with **dense vector retrieval** for automated candidate screening over unstructured resumes.
  - Owned architecture, implementation, and deployment as the **sole engineer**, collaborating directly with the founding team to integrate the pipeline into the production hiring workflow.

Technical Skills

Languages: Python, C++, SQL, JavaScript/TypeScript
GenAI / LLM: LLMs, RAG, GraphRAG, LangChain, LangGraph, FAISS, BM25, Neo4j, Ollama, Tavily, Prompt Engineering, Agentic AI, LLM Evaluation
Data Science / ML: Pandas, NumPy, scikit-learn, XGBoost, PyTorch, CNNs, RNNs/LSTMs, Transformers, Transfer Learning, Model Evaluation
Backend / Infrastructure: FastAPI, Django, REST APIs, PostgreSQL, Docker, Git, AWS (S3, Rekognition), Linux/WSL
Math: Probability, Statistics, Linear Algebra, Optimization, Stochastic Processes

Projects

- **Enterprise Knowledge Assistant** — *Python, LangGraph, LangChain, Neo4j, FAISS, BM25, FastAPI, Docker* Github
  - Built a **multi-agent enterprise search and decision-support system** with LangGraph, routing domain-specific queries across specialized HR, policy, precedent, and general-purpose agents.
  - Implemented **hybrid GraphRAG retrieval** combining **Neo4j knowledge graphs, FAISS dense retrieval, and BM25 lexical search** to capture both entity relationships and semantic relevance.
  - Added a **GenAI evaluation layer** to check response groundedness and citation accuracy, with traceable document IDs for source attribution.
  - Containerized the **FastAPI backend, Next.js frontend, and Neo4j database** using Docker Compose and added backend tests for graph-routing behavior.
- **Video Blog App** — *Django, RAG, AssemblyAI, Ollama, FFmpeg, yt-dlp* Github
  - Built a full-stack AI platform that automates the pipeline from **YouTube ingestion and audio extraction to transcription and long-form article generation** using Django, AssemblyAI, FFmpeg, yt-dlp, and Ollama.
  - Implemented **transcript-based RAG** for contextual question answering, semantic search, summarization, quiz generation, and interactive mind maps, grounding responses in source video content.
  - Engineered backend services for **media processing, transcript indexing, AI inference, and CRUD workflows**, with user authentication and persistent content management.
- **FlixPulse: Personalized Recommendation Engine** — *Python, FastAPI, React, SVD, KNN, Scikit-learn* Github
  - Developed an end-to-end recommendation system over **5.7M+ ratings from 21K+ users and 1.2K+ movies**, using SVD matrix factorization and item-based collaborative filtering.
  - Built a **FastAPI + React** product layer with cached recommendations, explainable item similarity, and popularity-based **cold-start fallbacks**; achieved **MAP@10 = 0.743** and **NDCG@10 = 0.834**.
  - Evaluated ranking quality using RMSE, MAE, MAP@10, NDCG@10, Hit Rate, Coverage, and Diversity, achieving **99.96% user coverage** on the benchmark dataset.

Achievements and Competitive Programming

- **AIR 1489**, JEE Advanced 2023 — top 0.05% among ∼180k candidates nationwide
- **AIR 841**, KVPY SA 2020 — National Science Fellowship (DST, Government of India)
- **Codeforces:** sisyphus_7 — Rated **Expert** (max rating 1607); solved problems across data structures, graph algorithms, and number theory